

Exercise 3: Boost Circuit

Introduction

The **Boost** converter of this experiment shown in figure 1. You need to build it using MATLAB SIMULINK.

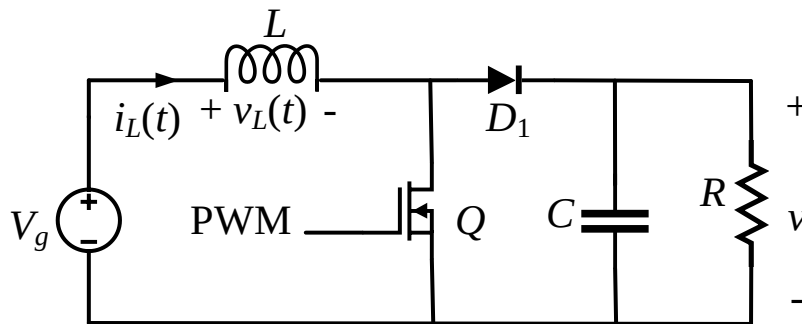


Figure 1

The **MOSFET** are well known for, provides the designer with an extremely efficient and reliable device for use in a wide variety of applications, configured to perform a turn-on and turn-off by a **PWM** signal. This **MOSFET** has an on-resistance $R_{on} = 80\text{m}\Omega$. The **DIODE** D_1 has an on-resistance $R_D = 100\text{m}\Omega$ and a constant forward voltage drop $V_D = 0.8\text{V}$. The value of **INDUCTOR** L is 1mH and it has a winding resistance $R_L = 0.05\Omega$, **CAPACITOR** C is $220\mu\text{F}/35\text{V}$ and R is a $1\text{k}\Omega/1\text{W}$ cement resistor.

The theoretical dc component of output voltage of the ideal Boost circuit is given by:

$$V = \frac{V_g}{1-D}$$

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Where D is the duty cycle, V_g is the input dc voltage.

Step 1: Build Boost Circuit

Configure the parameters for **Figure 1**.

Step 2: Configure the function generator of PWM

Set function generator to 20kHz square waveform signal and voltage source to 5V.

Measurements and observations

1. Create a table of results on your lab book for the **DUTY CYCLE** of frequency.
2. To fill the table, change the **DUTY CYCLE** of the input frequency from **10% to 90% (or less)**, and **REPEAT** your measurements, **RECORDING** and **PLOTTING** your results each time.
3. Observe and record the voltage ripple on the **OUTPUT** terminal when the **DUTY CYCLE** changes.
4. Observe and record the voltage and current waveform on the inductor L when the **DUTY CYCLE** changes.

5. Adjusting the power supply voltage to 2 volts, repeating 1 – 4.

Laboratory report (100/100 marks)

- Objective & background (total mark - 30/100)
- Calculations, graphs, and results (total mark - 40/100)
- Discussion of results and conclusions (total mark - 30/100)